

Linear Models 4: Tree growth Lab (Graphs)

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1 Getting data

We are going to use the *d.trees* data set again. This data set contains the growth rates of 557 trees.

```
d.trees <- read.csv2("../..//Datasets/TreesChamagne2017_Lab_modified.csv",
                     stringsAsFactors = TRUE)
##
str(d.trees)
```

```
'data.frame':  557 obs. of  12 variables:
 $ growth.rate      : num  0.702 1.139 1.394 1 1.355 ...
 $ species          : Factor w/ 4 levels "Beech","Larch",...: 1 1 1 4 4 4 1 1 1 1 ...
 $ site             : int   1 1 12 12 12 12 12 12 12 12 ...
 $ Density.tree.Class: Factor w/ 3 levels "high","low","medium": 1 2 3 3 1 1 2 3 2 3 ...
 $ age              : num   55 42 60 63 58 57 65 61 65 58 ...
 $ size             : num  23296 33718 57300 67297 97764 ...
 $ density.site      : num   43.6 43.6 35.8 35.8 35.8 ...
 $ density.tree      : num   6.03 3.91 5.54 5.06 5.89 ...
 $ diversity.tree     : num   1 1.14 1.98 2.67 2.19 ...
 $ diversity.site     : num   1.28 1.28 2.27 2.27 2.27 ...
 $ sp.richness        : int   1 1 2 2 2 2 2 2 2 2 ...
 $ SiteID            : int   1 1 12 12 12 12 12 12 12 12 ...
```

```
head(d.trees)
```

	growth.rate	species	site	Density.tree.Class	age	size	density.site
1	0.7	Beech	1	high	55	23296	44
2	1.1	Beech	1	low	42	33718	44
3	1.4	Beech	12	medium	60	57300	36
4	1.0	Spruce	12	medium	63	67297	36
5	1.4	Spruce	12	high	58	97764	36
6	1.2	Spruce	12	high	57	67373	36

	density.tree	diversity.tree	diversity.site	sp.richness	SiteID
1	6.0		1.0	1.3	1
2	3.9		1.1	1.3	1
3	5.5		2.0	2.3	12
4	5.1		2.7	2.3	12
5	5.9		2.2	2.3	12
6	10.2		2.0	2.3	12

2 Graphical analysis

In any data analysis it is usual practice that after having prepared the data, a graphical analysis is carried out. This step enables us to **inspect the data**, to **find mistakes** and to **better understand the data structure**.

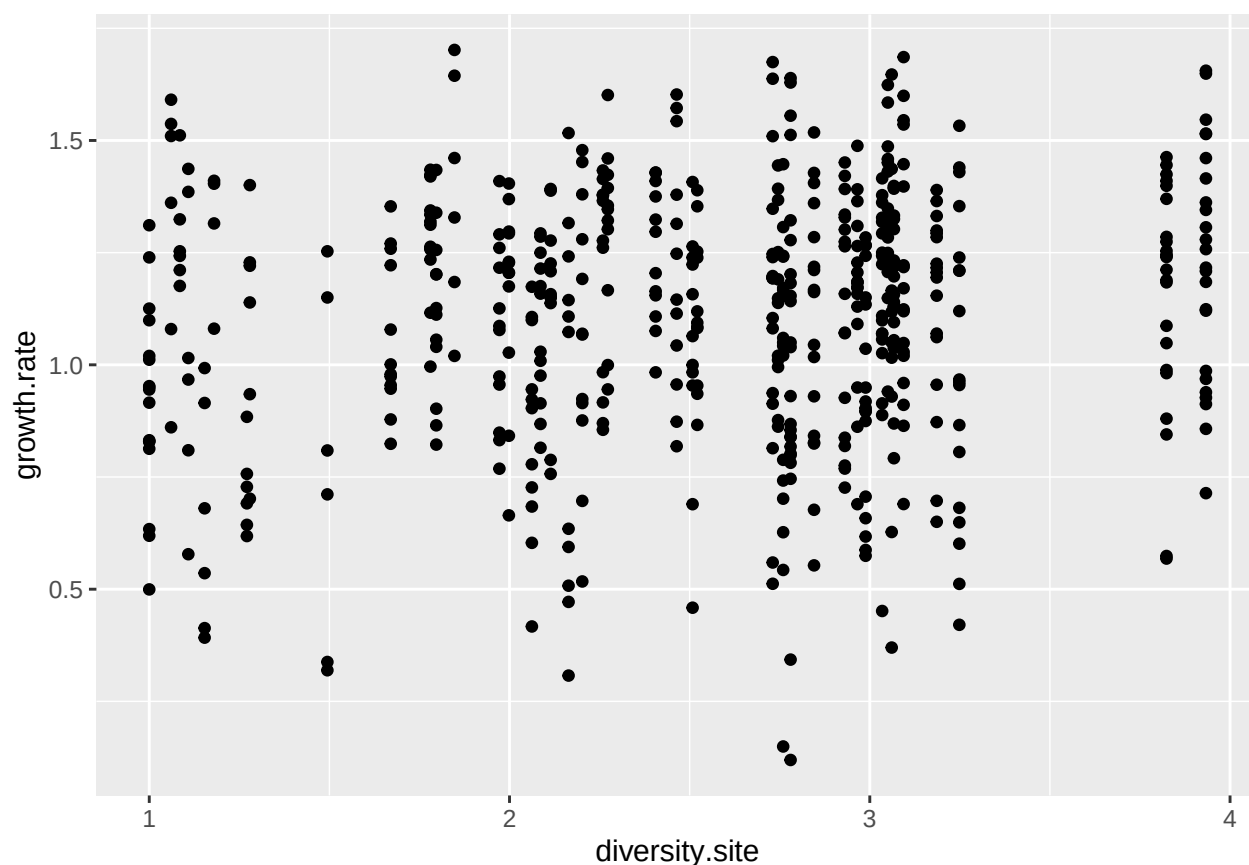
The usual way to carry out a graphical analysis, often also called Exploratory Data Analysis, is to plot the response variable against all predictors as well to produce other plots that may be needed.

2.1 Effect of diversity on growth

The first predictor that we inspect is the “site diversity”. This predictor quantifies the biodiversity of the site where a given tree is present. More diverse sites are expected to be associated with larger growth rates. In other words, the measured growth rates should positively correlate with the predictor *diversity.site*.

We start by plotting the response variable against *diversity.site*. Note that to produce graphs we use the add-on package *{ggplot2}*.

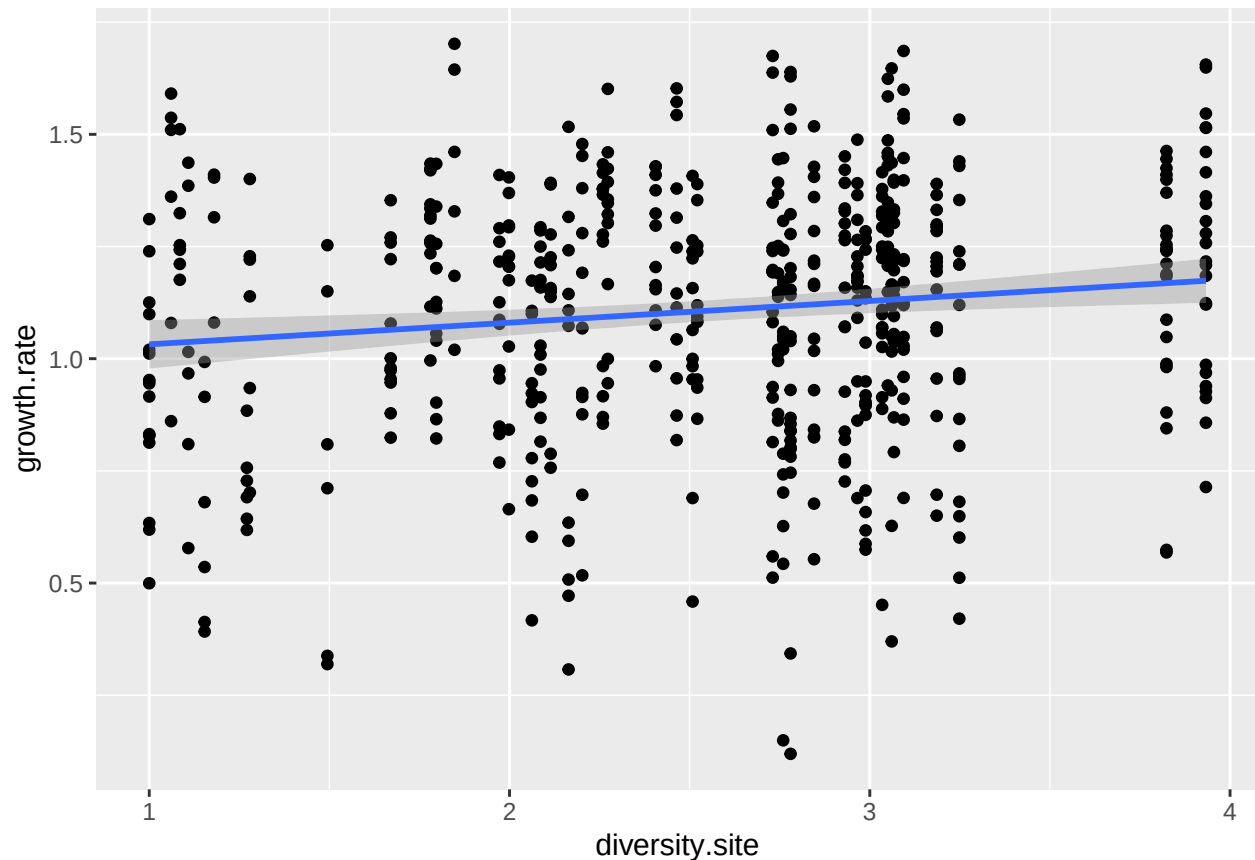
```
library(ggplot2)
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = diversity.site)) +
  geom_point()
```



The aim of these preliminary graphs is, among others, to better understand the data and to see what kind of relationship there is between the predictor and the response variable. As we will fit a regression model we may want to add a regression line to this plot.

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = diversity.site)) +
  geom_point() +
  geom_smooth(method = "lm")
```

‘geom_smooth()’ using formula = ‘y ~ x’



The regression line seems to be slightly positive, but very weak. Note that the regression line (in blue) is plotted with its confidence intervals (gray band).

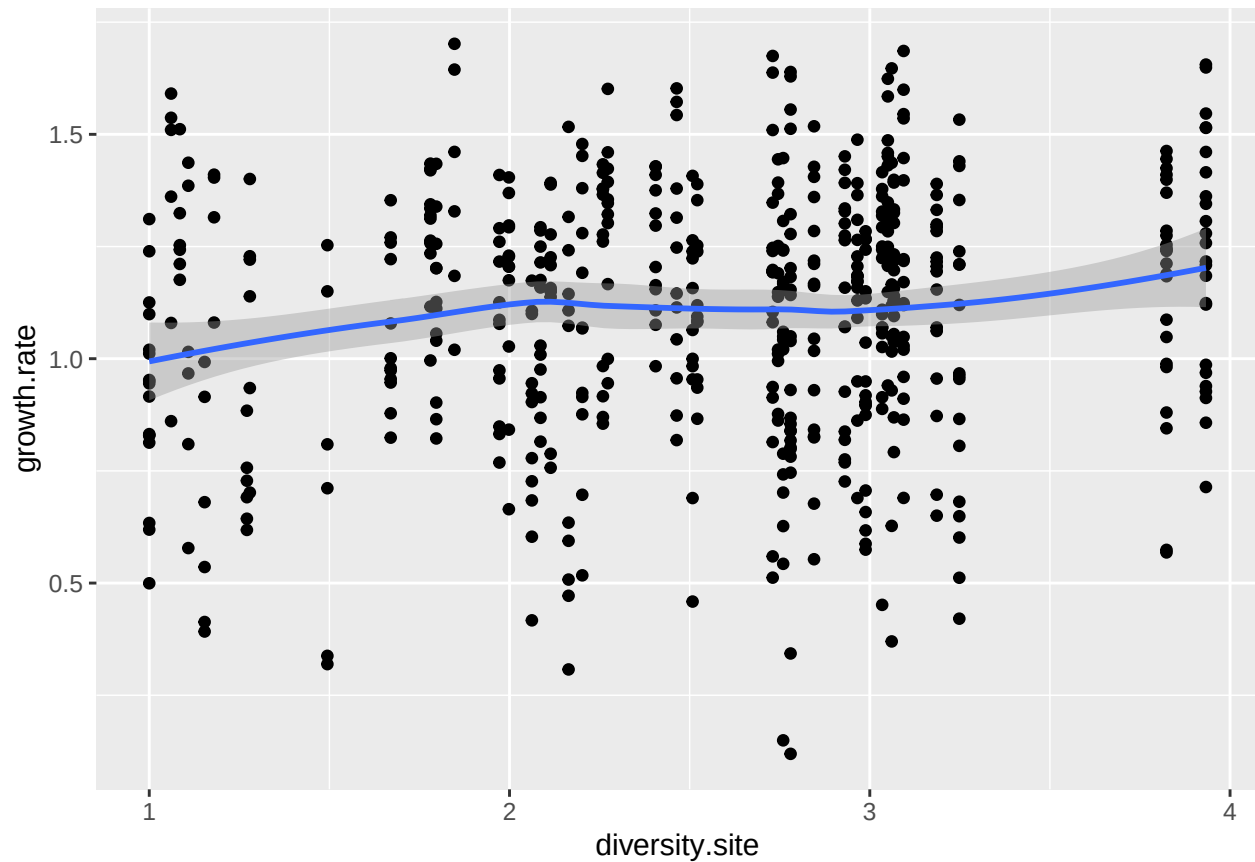
The aim of these preliminary plots is also to “pre-test” whether the assumptions that we will make when fitting the model are likely to be fulfilled. An assumption that is often made, but often not fulfilled in reality, is the linearity of the effects. For example, here we may want to include the predictor *diversity.site* into the model while assuming a linear effect (i.e. `lm(growth.rate ~ diversity.site)`). However, the effect of *diversity.site* on growth rates may be non-linear. **As we will see below, linear models can effectively model non-linear effects¹.**

So to “pre-test” the linearity assumption for this predictor we are adding a smoother rather than a regression line to the graph. A smoother can be seen as a local mean value that varies “smoothly” over the entire range of the *diversity.site* predictor (the next hour will indeed be dedicated to smoothing techniques)².

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = diversity.site)) +
  geom_point() +
  geom_smooth()
```

¹For “Linear Models” the adjective “Linear” refers to the linearity in the coefficients, not to the type of relationships modelled. This concept is often misunderstood.

²In time series analysis you may have seen the method called “moving average”, which produces similar results.

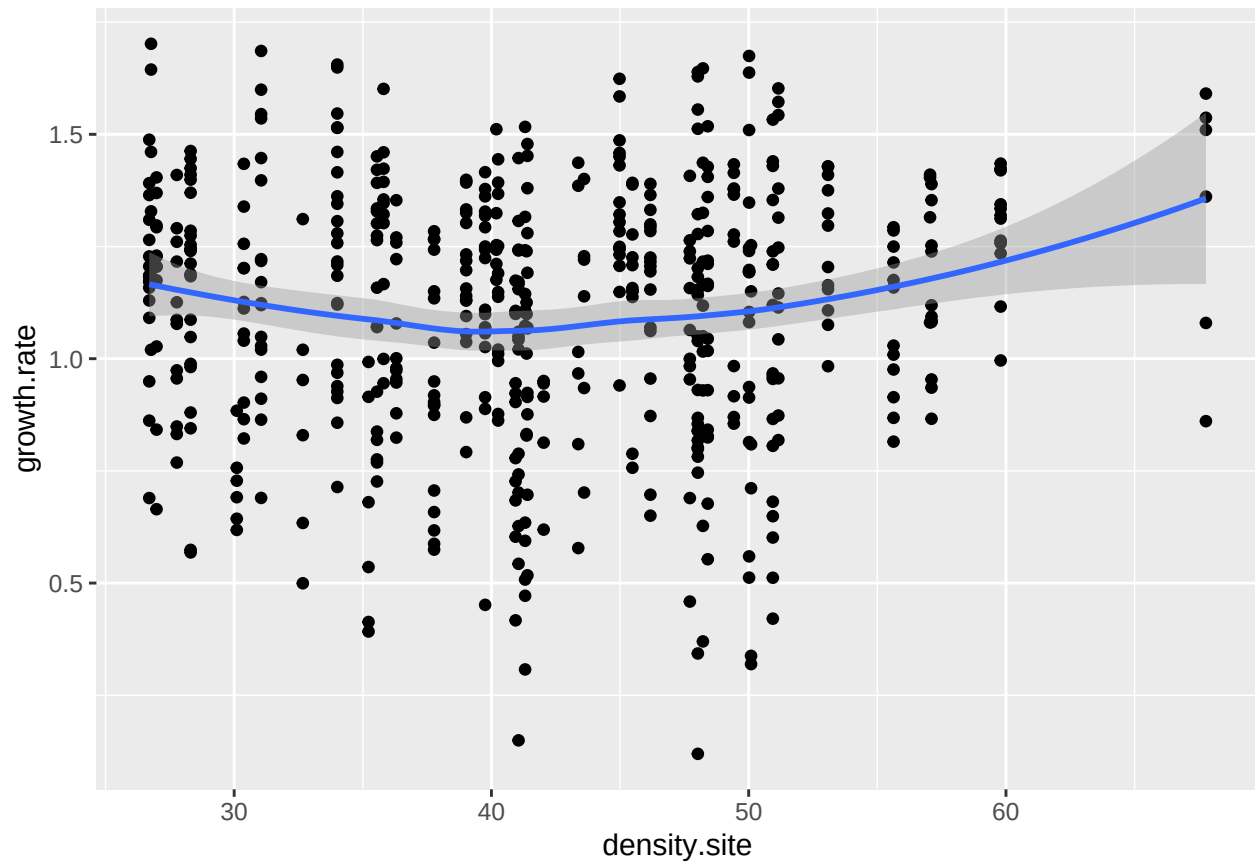


The smoother on this graph is not a perfect straight line, however, is close to it. Therefore, we may assume that there is no hint that the relationship between the response variable and the *diversity.site* predictor may be non-linear. Therefore, we may start including this variable in the model as a linear effect.

2.2 Effect of density on growth

Let's visualise the effect of tree density (measured at site level) on growth rates.

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = density.site)) +
  geom_point() +
  geom_smooth()
```

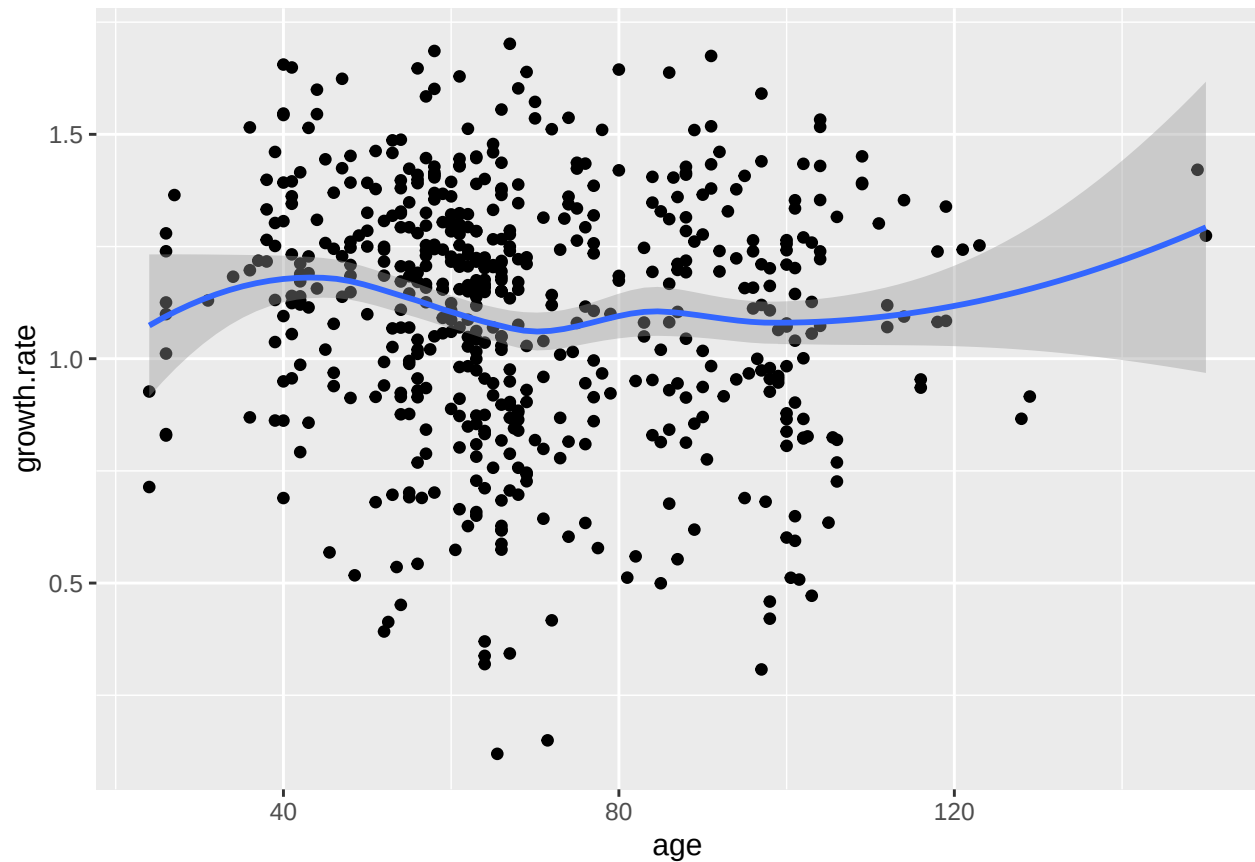


This effect is clearly non-linear. The smoother indicates that we may model this relationship as a quadratic effect.

2.3 Effect of age on growth

Let's visualise the effect of tree age on the growth rates.

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = age)) +
  geom_point() +
  geom_smooth()
```

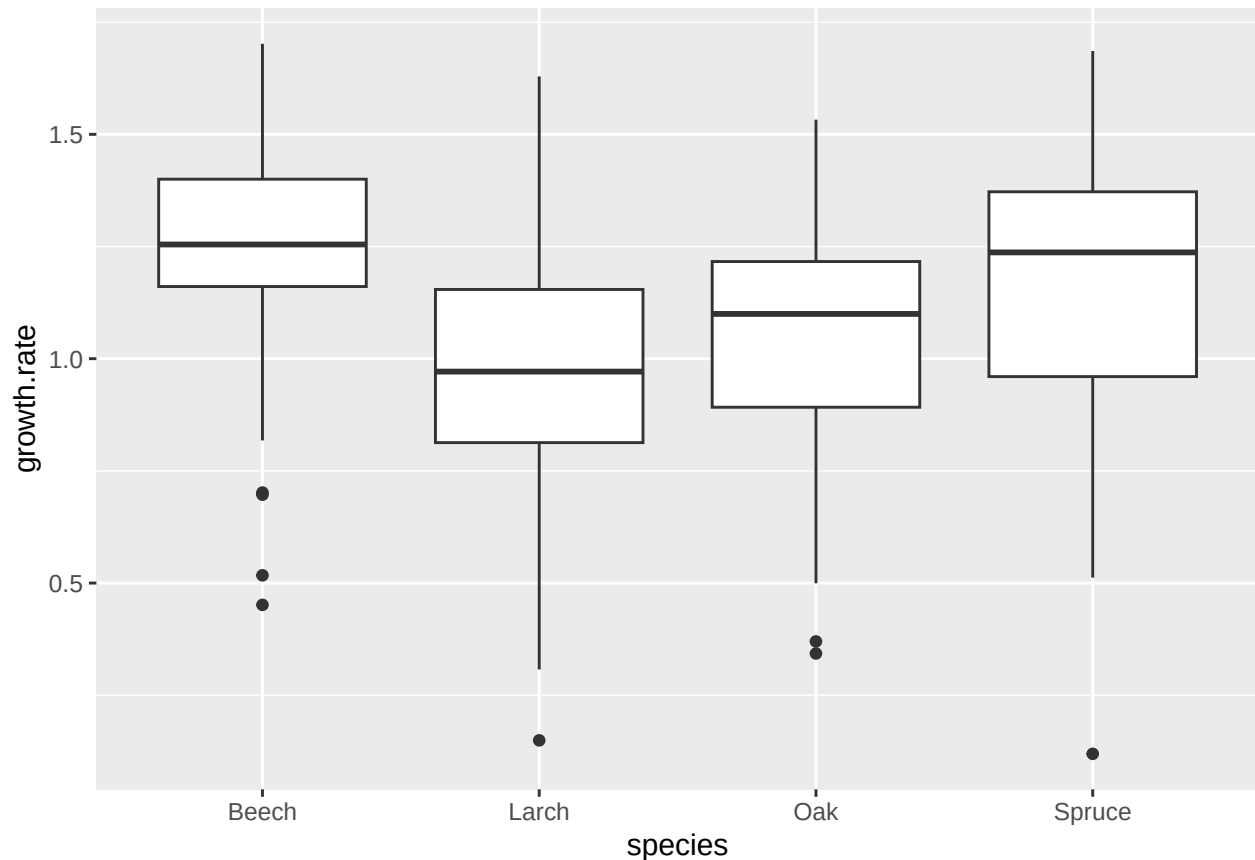


We don't see any clear effect of this predictor. The smoother is not straight, but the observed oscillations, may be due to randomness. We may want to include this variable into the starting model as a linear effect. Note that in general some field knowledge and experience is required to be able to interpret these kind of graphs.

2.4 Effect of species on growth

Let's visualise the effect of tree species on the growth rates. As *species* is a categorical variable, we do not use a scatterplot, but rather a boxplot.

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = species)) +
  geom_boxplot()
```



Based on biological knowledge, the predictor *species* is expected to be a very important predictor and this is confirmed by this graph.

Note that the ordering of these graphs is based on didactic reasons not on the importance of the predictors. There are no strict rules on which predictors should be visualised first.

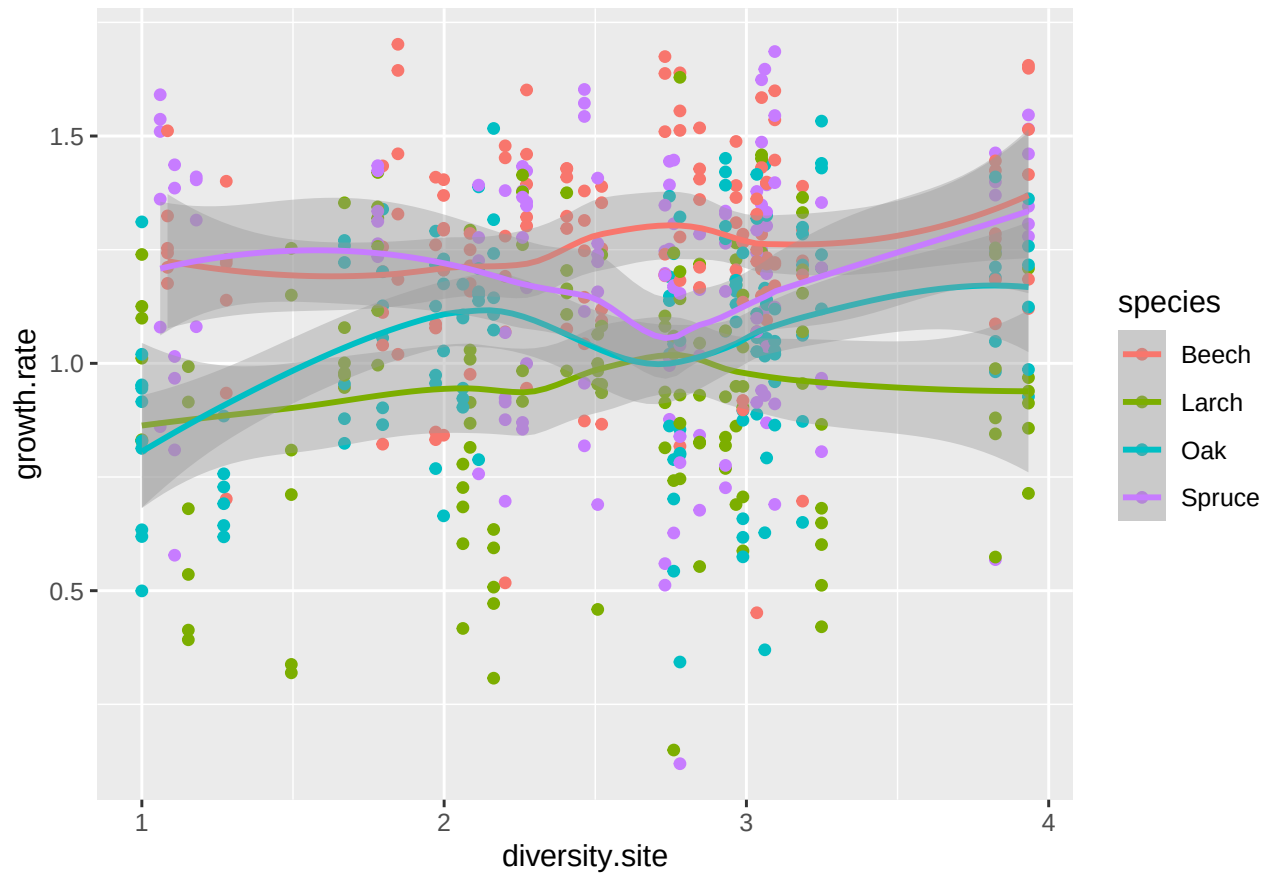
2.5 Further “case-specific” plots

Each data analysis has its own peculiarities that must be considered when analysing the data. For example here we expected the effect of all the continuous predictors to differ among species. In terms of modelling this may be accounted with interactions terms.

To “pre-test” the hypothesis that *diversity.site*, *density.site* and *age* interact with *species* we create the corresponding plots.

Let’s inspect graphically, whether *diversity.site* interacts with *species*. There are two ways to do that. Let’s start with “grouping” observations of the same species by using different colours.

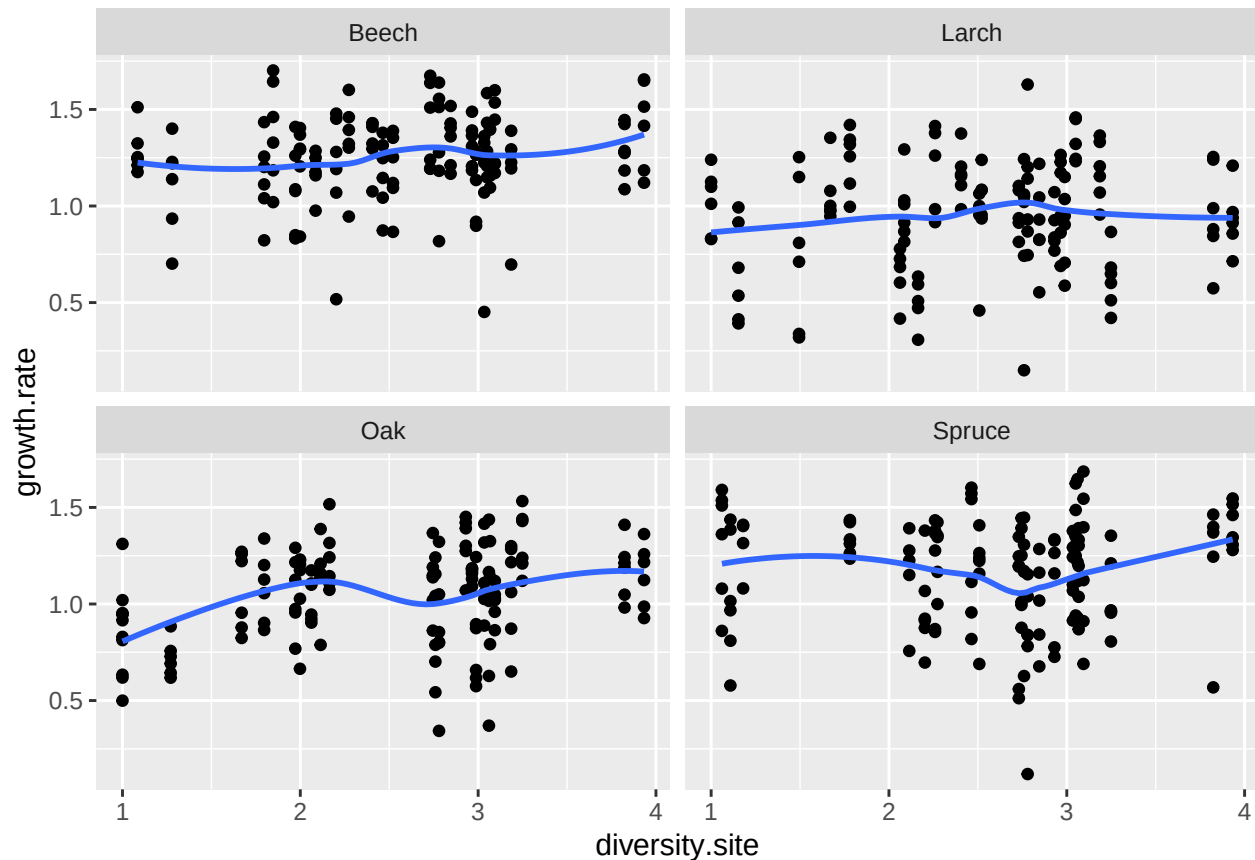
```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = diversity.site,
                     colour = species)) + ## ! new argument
  geom_point() +
  geom_smooth()
```

There is no clear indication that the effect of *diversity.site* strongly differs among species. They all seem to be weak and almost flat on zero.

Note, that this graph is “quite full”. It is somehow difficult to take apart all pieces of information. The better approach to visualise the effect of *diversity.site* for each species separately is to use “panelling”. This means that the observation of each species will be plotted in a different panel.

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = diversity.site)) +
  geom_point() +
  geom_smooth(se = FALSE) +
  facet_wrap(~ species) ## ! new argument
```

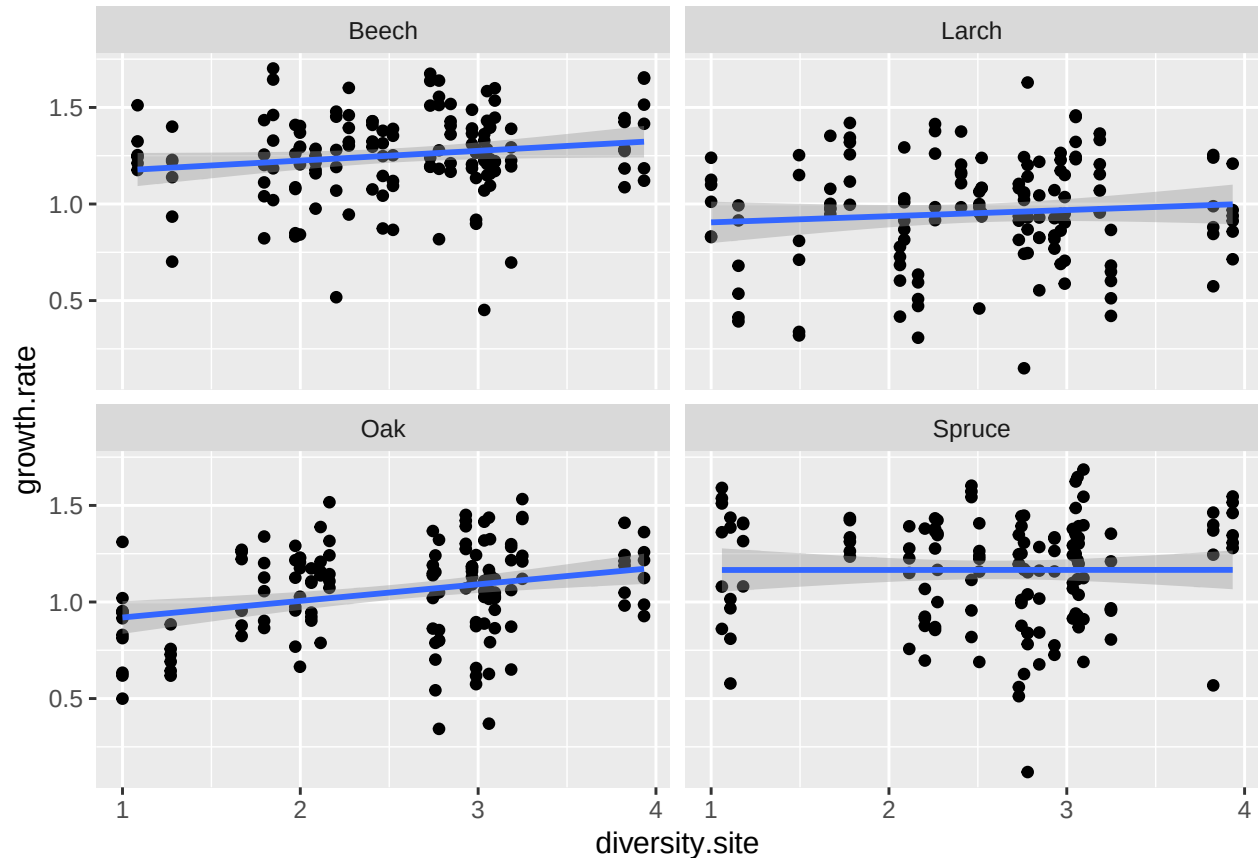


This graph, that is more easily readable, confirms that diversity does not seem to have a very strong effect on growth rates. Indeed, most of the smoothers are fairly flat. As there are no clear difference among smoothers, we may expect the interaction between *species* and *diversity.site* to be unneeded.

Note that smoothers may be unnecessarily wiggly when fitted to small sample size. Indeed, some of the non-linearities seen above, may be due to noise naturally present in the data (i.e. a few observations with large error). In this very case, we may want to revert back to regression lines.

```
ggplot(data = d.trees,
       mapping = aes(y = growth.rate,
                     x = diversity.site)) +
  geom_point() +
  geom_smooth(method = "lm") +
  facet_wrap(. ~ species)
```

‘geom_smooth()’ using formula = ‘y ~ x’



Once again, we confirm that *diversity.site* seem to have a weak effect that does not clearly differ among species.

One important point to recognise here, is that a graph is always built in a progressive way. We don't start with a super complex graph, but rather start simple and then add elements to it. This way of proceeding does not imply that we must keep all steps needed to get to the final plot in our document. Indeed, in most cases only the most informative graphs are retained.

For the sake of brevity, we omit the analysis for species-density and species-age interactions.

2.6 Notes on the graphical analysis

- The exploratory graphical analysis is also very useful to spot mistakes in the data.
- There is a trade-off between amount of information you can put on graphs and its readability.
- For example, on the very last graph above we visualised three sources of information on one plot (i.e. growth rate, diversity and species). Adding a fourth source of information (e.g. with colours) may lead to a unsatisfactory graph.
- Note that graphs for design variables such as sites are usually also needed. Further graphs may be maps for geo-located observations or time series plots for historical observations.

3 Modelling

3.1 Fitting the starting model

Model fitting is an iterative process. A starting model is fitted and then, based on the model diagnostics and the results interpretation, a better model is fitted.

The starting model must contain all predictors and all the corresponding interactions that we consider to be important. In this case a meaningful starting model could be.

```
lm.trees.0 <- lm(growth.rate ~ species +
  age + density.site + diversity.site +
  species:age +
  species:density.site +
  species:diversity.site,
  data = d.trees)
```

Note that in this model, we assume the effect of all continuous variables to be linear. Among all possible interactions, we allowed the model to contain only three two-way interactions.

3.2 Model “development”

After having fitted the starting model, we check whether all interactions are needed. In addition, we check whether we need to add non-linearities to the model. Note non-linearities (e.g. quadratic terms), could have been included in the starting model. There is no “right way” to develop a model. Every case, is different and may need different solution. There is also an high degree of subjectivity in deciding what solution is the best.

Here we subjectively decided to start with a model with all main effects, all the interactions with *species*. Note also that we assumed all predictors to have a linear effect.

As a next step, we may want to test whether all three two-fold interactions are needed.

```
drop1(lm.trees.0, test = "F")
```

Single term deletions

Model:

```
growth.rate ~ species + age + density.site + diversity.site +
  species:age + species:density.site + species:diversity.site
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			33.3	-1536		
species:age	3	1.800	35.1	-1513	9.73	2.9e-06 ***
species:density.site	3	0.957	34.3	-1527	5.18	0.0016 **
species:diversity.site	3	0.466	33.8	-1535	2.52	0.0571 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The interaction between *species* and *age* is very strong. There is also strong evidence that the effect of density is different depending on tree species. Finally, is not entirely clear whether “diversity” interacts with *species*.

We may want to drop this interaction and refit the model again.

```
lm.trees.1 <- update(lm.trees.0, . ~ . - species:diversity.site)
drop1(lm.trees.1, test = "F")
```

Single term deletions

Model:

```
growth.rate ~ species + age + density.site + diversity.site +
  species:age + species:diversity.site
```

	Df	Sum of Sq	RSS	AIC	F value	Pr(>F)
<none>			33.8	-1535		
diversity.site	1	1.30	35.1	-1516	20.86	6.1e-06 ***
species:age	3	1.48	35.3	-1517	7.91	3.6e-05 ***
species:diversity.site	3	1.00	34.8	-1524	5.38	0.0012 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

All the terms present in the model are now highly significant and we could decide to and retain `lm.trees.1` as our final model.

Note that there are very many approaches to model building (or model development). They mainly differ in terms of:

- how are models compared (F-test, AIC, BIC, R^2 , ...)
- are models compared in pairs in a sequential way (as done above) or are all models compared simultaneously?
- how do we set the upper limit in complexity for our models? Above we only allowed three two-fold interactions and assumed all continuous variables to have a linear effect. Shall we include higher-term interactions or non-linear effects?
- ...

4 Session Information

```
sessionInfo()
```

R version 4.5.1 (2025-06-13)

Platform: aarch64-apple-darwin20

Running under: macOS Sequoia 15.6.1

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK vers.

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Zurich

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] ggplot2_3.5.2 knitr_1.50

loaded via a namespace (and not attached):

[1] vctrs_0.6.5	nlme_3.1-168	cli_3.6.5	rlang_1.1.6
[5] xfun_0.52	generics_0.1.4	labeling_0.4.3	glue_1.8.0
[9] htmltools_0.5.8.1	tinytex_0.57	scales_1.4.0	rmarkdown_2.29
[13] grid_4.5.1	evaluate_1.0.4	tibble_3.3.0	fastmap_1.2.0
[17] yaml_2.3.10	lifecycle_1.0.4	compiler_4.5.1	dplyr_1.1.4
[21] RColorBrewer_1.1-3	pkgconfig_2.0.3	mgcv_1.9-3	rstudioapi_0.17.1
[25] lattice_0.22-7	farver_2.1.2	digest_0.6.37	R6_2.6.1
[29] tidyselect_1.2.1	splines_4.5.1	pillar_1.11.0	magrittr_2.0.3
[33] Matrix_1.7-3	withr_3.0.2	tools_4.5.1	gtable_0.3.6